

TUT - 3:

T3.1: $p(x) = \sum_{i=1}^K a_i x^i$; $q(x) = \sum_{i=1}^L b_i x^i$

$\mathbb{Z}\mathbb{Z}$: $\lim_{n \rightarrow \infty} \frac{p(n)}{q(n)} = \begin{cases} 0 & , K < L \\ \frac{a_K}{b_L} & , K = L \\ \pm \infty & , K > L, \text{ je nach } \sigma\left(\frac{a_K}{b_L}\right) \end{cases}$

Bew: Wir klammern jeweils die höchste Potenz aus:

$$\frac{p(n)}{q(n)} = \frac{a_K n^K \left(\sum_{i=1}^K \frac{a_i n^i}{a_K n^K} \right)}{b_L n^L \left(\sum_{i=1}^L \frac{b_i n^i}{b_L n^L} \right)} \xrightarrow{(*)} \frac{1}{1} = 1$$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{q(n)} = \lim_{n \rightarrow \infty} \frac{a_K}{b_L} \cdot n^{K-L}$$

$$\begin{cases} 0 & , K < L \\ \frac{a_K}{b_L} & , K = L \\ \pm \infty & , K > L, \sigma\left(\frac{a_K}{b_L}\right) \end{cases}$$

(*) $a_K n^K \left(\frac{a_1}{a_K} \frac{1}{n^{K-1}} + \frac{a_2}{a_K} \frac{1}{n^{K-2}} + \dots + \frac{a_K}{a_K} \frac{1}{n^{K-K}} \right) \rightarrow 1$

exp. abfallender fkt.

$$\frac{n}{n^{K+1}}$$

(b) $\mathbb{Z}\mathbb{Z}$: $K \in \mathbb{N}_0, |a| < 1, \boxed{n^K} \boxed{q^n} \xrightarrow{n \rightarrow \infty} 0$
 ↑ polynomische wachsen der fkt.

Bew: $q = \frac{1}{1+x}, x > 0$

$n > K$

Abschätzung: $(1+x)^n = \sum_{i=0}^n \binom{n}{i} x^i \geq \binom{n}{K+1} x^{K+1}$

$$\binom{n}{K+1} = \frac{n!}{(n-K-1)!(K+1)!} = \frac{n \cdot (n-1) \cdot \dots \cdot (n-K)}{(K+1)!} = \prod_{i=0}^K \frac{n-i}{(K+1)!}$$

$\forall i \in \{0, \dots, K\}: n-i \geq \frac{n}{K+1} \quad \forall (n > K)$

$$\binom{n}{K+1} \geq \frac{1}{(K+1)!} \left(\frac{n}{K+1} \right)^{K+1} n - K \geq \frac{n}{K+1} \Rightarrow n + nK - K - K^2 \geq n$$

$$\Leftrightarrow (1+x)^n \geq \frac{1}{(K+1)!} \left(\frac{n}{K+1} \right)^{K+1} \cdot x^{K+1}$$

$$0 \leq |n^k q^n| \leq \frac{n^k}{(k+1)! \left(\frac{n}{k+1}\right)^{k+1} \cdot x^{k+1}} \sim \frac{n^k}{n^{k+1}} \xrightarrow{a_1} 0$$

(c) zz: $a > 0 : \sqrt[k]{a} \xrightarrow{n \rightarrow \infty} 1$

Bew. I. ($a=1$): $\lim_{n \rightarrow \infty} \sqrt[n]{1} = 1, x^n = 1^{1+n} = 1 \forall n.$

II. ($a > 1$), $x_n = 1 + b_n$, $b_n > 0$ T2. -
Bernoulli (b)
ungl.

$$x_n = \sqrt[n]{a} = 1 + b_n \xrightarrow{(\cdot)^n} a = (1 + b_n)^n \geq 1 + n \cdot b_n$$

$$\frac{a-1}{n} \geq b_n > 0 \Rightarrow \lim_{n \rightarrow \infty} \frac{a-1}{n} \geq \lim_{n \rightarrow \infty} b_n > 0$$

$$b_n \xrightarrow{n \rightarrow \infty} 0 \Leftrightarrow 0 \geq \lim_{n \rightarrow \infty} b_n > 0$$

$$x_n \xrightarrow{n \rightarrow \infty} 1, x_n = 1 + b_n \Leftrightarrow \lim_{n \rightarrow \infty} x_n = 1 + \lim_{n \rightarrow \infty} b_n = 1 \quad \square$$

III. $0 < a < 1$

Sei $b := \frac{1}{a}, b > 1$ $x_n = 1 - b_n, b_n > 0$

(II) $\Leftrightarrow \lim_{n \rightarrow \infty} \sqrt[n]{b} = 1 = \lim_{n \rightarrow \infty} \left(\frac{1}{a}\right)^{1/n}$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1^{1/n}}{a^{1/n}} = \frac{1}{\underbrace{\lim_{n \rightarrow \infty} a^{1/n}}_{=: L}} = 1$$

$\Leftrightarrow \frac{1}{L} = 1 \Leftrightarrow \boxed{1 = L} \quad \square$

$$b_n = \sqrt[k]{a_n} \Leftrightarrow b_n^k = a_n$$

$$\lim_{n \rightarrow \infty} b_n^k = \lim_{n \rightarrow \infty} a_n$$

$$\Leftrightarrow \left(\lim_{n \rightarrow \infty} b_n\right)^k = \lim_{n \rightarrow \infty} a_n$$

$$\left(\lim_{n \rightarrow \infty} b_n\right)^k = a$$

$$\Leftrightarrow \boxed{\lim_{n \rightarrow \infty} b_n = \sqrt[k]{a}}$$

T. 3.2: $(a_n)_{n \in \mathbb{N}} \subset [0, \infty), a_n \xrightarrow{n \rightarrow \infty} a$

zz: $\sqrt[n]{a_n} \rightarrow \sqrt[k]{a}$

(w) geg: $a_n \leq a \quad \forall n \in \mathbb{N}$, monoton

Bew.: $b_n := \sqrt[n]{a_n} \leq \sqrt[n]{a}$ monoton + n.o. beschränkt.

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \sqrt[n]{a_n} \text{ existiert.}$$

T 3.2. (b):

geg: $a \geq 0$

geg: $a \geq 0 \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$

Bew: $\forall \varepsilon > 0 \exists N \in \mathbb{N} \quad n > N : 0 \leq |a_n - 0| < \varepsilon$

wähle z.B. $\tilde{\varepsilon} = \sqrt{\varepsilon} > 0$

$$0 \leq \sqrt[n]{a} < \tilde{\varepsilon}$$

$$0 \leq |\sqrt[n]{a_n} - 0| < \tilde{\varepsilon} \Leftrightarrow \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \sqrt[n]{a}$$

(c) geg:

geg: $a > 0, \quad \kappa = 2$

Bew: $\lim_{n \rightarrow \infty} a_n = a, \exists N \in \mathbb{N} \quad \forall n > N \quad a_n > \frac{a}{2} > 0$

$$\sqrt{a_n} \geq \sqrt{a/2} \Leftrightarrow \sqrt{a_n} + \sqrt{a} \geq \sqrt{a/2} + \sqrt{a}$$

$$0 < \left| \sqrt{a_n} - \sqrt{a} \right| < \frac{a_n - a}{\sqrt{a/2} + \sqrt{a}}$$

$$\lim_{n \rightarrow \infty} a_n = a \Leftrightarrow \lim_{n \rightarrow \infty} a_n - a = 0$$

$$0 < \lim_{n \rightarrow \infty} (\sqrt{a_n} - \sqrt{a}) < 0$$

$$\Rightarrow \text{Sandwiche theorem} \rightarrow \lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a} \quad \square$$

$$\left(\sqrt{a_n} + \sqrt{a} \cdot (\sqrt{a_n} - \sqrt{a}) = \frac{a_n - a}{\sqrt{a_n} - \sqrt{a}} \right)$$

T 3.3:

(c) $\lim_{n \rightarrow \infty} \frac{\sqrt{n^2+1} - \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$

$$= \lim_{n \rightarrow \infty} \frac{(n^2+1) - n^2}{\sqrt{n+1} + \sqrt{n}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2\sqrt{n}} = 0$$

(e) $\lim_{n \rightarrow \infty} \frac{n}{n(\sqrt{1+1/n} + 1)} = \frac{1}{2}$

(f) $\lim_{n \rightarrow \infty} n^7 \cdot \left(\frac{1}{2}\right)^n \rightarrow 0$

$$\lim_{n \rightarrow \infty} \frac{2^n}{n^7} = \infty$$